

Macro I

NAME:

Problem Set 2

Submit Questions 1 and 2

Questions 3 and 4 will be discussed during TA sessions

Questions 5-6 are for extra practice and include solutions

Question 1: A mini life cycle model

Assume that a consumer lives for 3 periods: 0, 1, and 2. The consumer's income streams are $y_0 = 100, y_1 = 300, y_2 = 25$. Assume that the real interest rate is constant at 4%, and that $\beta = \frac{1}{1+r}$. The consumer periodic utility functions is logarithmic, so that:

$$U(c_0, c_1, c_2) = \ln(c_0) + \beta \ln(c_1) + \beta^2 \ln(c_2)$$

Answer the following questions. Provide brief explanations and/or show your work.

1. Briefly discuss - what sort of life cycle evolution can these assumptions correspond to?
2. Do the consumer want to consume equal quantities every period? Why or why not?
3. What is the consumer's PVLR?
4. Calculate the consumer's optimal consumption plan (i.e. how much consumption in every period).
5. Calculate the consumer's optimal saving in every period.
6. How does the consumer's wealth evolves over his/her life?
7. Suppose that another consumer faces the same income process and interest rate, but has a different discount factor: $\beta = 0.9711$. Solve for the optimal consumption plan for the second consumer. Briefly explain the difference.

Question 2: Optimal Consumption Plan with log utility

Assume that:

- consumers live for two periods
- discount the future with a discount factor $0 < \beta < 1$
- income in the first period is y_0 ; income in the second period is y_1 ; no initial assets $a_0 = 0$.
- the utility function in each period is $u(c) = \ln(c)$
- the real interest rate is some positive constant $r > 0$

As a results, the objective function is to choose c_0 and c_1 in order to maximize the discounted sum of utilities:

$$\max_{c_0, c_1} \{\ln(c_0) + \beta \ln(c_1)\}$$

Answer the following questions. Provide brief explanations and/or show your work.

1. What is the lifetime budget constraint?
2. Write the Lagrangian for this problem.
3. Derive the first order conditions
4. What is the Euler equation for this problem? Use it to express c_1 as a function of r, β, c_0
5. Substitute your answer for c_1 into the budget constraint and use it to solve for c_0 , and then for c_1
6. Derive a term for savings in the first period as a function of y_0, y_1, r, β .
7. Consider each of the following changes in isolation (i.e. keep everything else constant and change just one thing). For each, will the consumer save more or less? Show using the solution to the previous part and explain why.
 - (a) Higher β
 - (b) Higher y_0
 - (c) Higher y_1

Question 3 - For TA Session - CRRA utility function, income and substitution effects

The purpose of this question is to: (i) have another practice at deriving the optimal consumption plan; (ii) illustrate the potentially competing income and substitution effects that we discussed in class.

One of the most widely used utility functions in macroeconomic research is:

$$u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}, \quad \sigma > 0$$

(This is usually called either the constant relative risk aversion (CRRA) utility function or the iso-elastic utility function).

Assume that:

- consumers live for two periods
- discount the future with a discount factor $0 < \beta < 1$
- income in the first period is y_0 ; income in the second period is y_1 ; no initial assets $a_0 = 0$.
- the utility function in each period is the CRRA utility function

- the real interest rate is some positive constant $r > 0$

Answer the following:

1. What is the lifetime budget constraint?
2. Write the Lagrangian for this problem.
3. Derive the first order conditions
4. What is the Euler equation for this problem? Use it to express c_1 as a function of r, β, σ, c_0
5. Assume that $y_0 = 0$ and $y_1 > 0$ (so that the consumer is necessarily a borrower). Substitute your answer for c_1 into the budget constraint and use it to solve for c_0 . Use your solution to show that in this case there is no ambiguity and that a higher r will definitely lower current consumption.
6. Now assume that $y_0 > 0$ and $y_1 = 0$ (so that the consumer is necessarily a lender). Substitute your answer for c_1 from the Euler equation into the budget constraint and use it to solve for c_0 . Use your solution to show that in this case there is ambiguity and that the effect of r on current consumption depends on the value of σ . Briefly discuss the competing effects. What should we assume about σ in order to be consistent with the assumption we made in class about the effect of r on current consumption?

Question 4 - For TA session

Use the predictions of the permanent income hypothesis that we discussed in class to rank the following from the biggest to the smallest consumption increase among recipients. (For simplicity, you may assume that the desired consumption plan is spending equal amounts over the life cycle). Explain your answer.

- (a) an unexpected and explicitly temporary tax rebate of \$300 (assume the consumer believes that he/she will never have to pay this back to the government)
- (b) a special dividend on a stock that the consumer owns of \$300 per shareholder, that also lowers the value of the stock by \$300 per share (assume the consumer owns one stock).
- (c) an unexpected raise of \$300 per year, effective immediately

Question 5 - Extra / For TA session

Assume that consumers behave according to the assumptions of the consumption model described in class. Assume that you collected data on their consumption and the real interest rate. You may also assume that all consumers share the same preferences and the same expectations regarding their current and future income.

1. In your data, you observe that some consumers increased their consumption when the real interest rate increased. Can you say with certainty whether these consumers were lenders/borrowers/neither?
2. Other consumers decreased their consumption when the real interest rate increased. Can you say with certainty whether these consumers were lenders/borrowers/neither?

Solution:

In general, an increase in r triggers both income and substitution effects. The substitution effect, captured by the Euler equation, takes into account the fact that current consumption becomes relatively more expensive, and therefore consumers would like to reduce their current consumption. Using the Euler equation we can start with the assumption that it was holding with equality before the change, and now $u'(c) < \beta(1+r)u'(c^f)$. Therefore, in order to re-optimize, we have to lower c , which will increase $u'(c)$ and increase c^f , which will lower $u'(c^f)$.

In addition, there is an income effect: if a consumer is a lender, then the higher interest rate acts like a positive income shock, triggering a force that increases current consumption. However, if the consumer is a borrower, then the higher r triggers a negative income shock, lowering the demand for current consumption.

With this reasoning we can conclude that:

1. If current consumption increased then it must be that the consumer is a lender, because this is the only way that demand for consumption will rise.
2. If current consumption falls we can't tell. The reason is that consumption may fall because a consumer is a borrower, or because the consumer is a lender and the substitution effect is stronger than the income effect.

Question 6 - Extra / For TA session

Assume that there are two consumers, Jerry and George. Both live for two periods, are price takers, and have identical utility function: $U(c_i, c_i^f) = \ln(c_i) + \beta \ln(c_i^f)$ where $0 < \beta < 1$, and i can be J for Jerry, or G for George.

Both consumers have initial wealth $a = 0$. The endowment (or income) process is different:

$$\text{Jerry:} \quad y_J = y_0 \quad y_J^f = y_1$$

$$\text{George:} \quad y_G = \frac{y_1}{1+r} \quad y_G^f = (1+r)y_0$$

where we assume that $y_0 > \frac{y_1}{\beta(1+r)}$

- (a) Write the optimization problem for each consumer. Use the lifetime budget constraint.
- (b) Derive the Euler equation
- (c) Use the Euler equation and the lifetime budget constraint to solve for Jerry's and George's optimal consumption plans
 - (i) Are the optimal consumption plans different for the two consumers? Explain why or why not.
 - (ii) Do Jerry and George engage in borrowing or lending in the first period? calculate the amounts and explain who is borrowing and who is lending, and why.
- (d) Now assume that the economy changes, and borrowing is no longer allowed. (To avoid unnecessary complications, you may assume that if someone wishes to lend, there is always an opportunity to do so for a rate r , e.g. to foreigners). **NO CALCULATIONS ARE REQUIRED FOR THE FOLLOWING QUESTIONS, JUST EXPLAIN.**
 - (i) Does the Euler equation that you derived in part (b) still hold for both consumers? for none? for one? explain.
 - (ii) Assume that the government would like give a tax rebate in order to increase **current** aggregate consumption. Is it better to give the rebate to Jerry or to George? Explain.

Solution:

- (a) The two problems:

$$\begin{aligned} \max_{c_J, c_J^f} & \ln(c_J) + \beta \ln(c_J^f) \\ \text{s.t.:} & c_J + \frac{c_J^f}{1+r} = y_J + \frac{y_J^f}{1+r} = y_0 + \frac{y_1}{1+r} \end{aligned}$$

$$\begin{aligned} \max_{c_G, c_G^f} & \ln(c_G) + \beta \ln(c_G^f) \\ \text{s.t.:} & c_G + \frac{c_G^f}{1+r} = y_G + \frac{y_G^f}{1+r} = \frac{y_1}{1+r} + \frac{(1+r)y_0}{1+r} = y_0 + \frac{y_1}{1+r} \end{aligned}$$

- (b) Euler equation (this skips setting up the Lagrangian):

$$u'(c) = \beta(1+r)u'(c^f) \quad \Rightarrow \quad \frac{1}{c} = \beta(1+r)\frac{1}{c^f}$$

- (c) from the Euler equation we have: $c^f = \beta(1+r)c$. Substitute in the budget constraint:

$$c + \frac{\beta(1+r)c}{1+r} = y_0 + \frac{y_1}{1+r} \quad \Rightarrow \quad c = \frac{1}{1+\beta} \left[y_0 + \frac{y_1}{1+r} \right] \quad \Rightarrow \quad c^f = \beta(1+r)c = \frac{\beta}{1+\beta} [y_0(1+r) + y_1]$$

(c)(i) The two consumers have identical consumption plans because they have the same PVLR, and no restrictions on borrowing and lending, so they can smooth consumption optimally, as dictated by the Euler equation.

(c)(ii) While PVLR is identical, the timing of income is different. Specifically, George receives more of his income in the later period, which makes him the likely borrower in this case. Jerry receives more of his income in the early period, which makes him the likely saver/lender. To see this more accurately, compare consumption to income in the current period:

$$c_J - y_J = \frac{1}{1+\beta} \left[y_0 + \frac{y_1}{1+r} \right] - y_0 = -\frac{\beta}{1+\beta} y_0 + \frac{y_1}{(1+\beta)(1+r)}$$

Note that our assumption that $y_0 > \frac{y_1}{\beta(1+r)}$ implies that the difference between consumption and income for Jerry is negative, and so Jerry is a lender.

Similarly,

$$\begin{aligned} c_G - y_G &= \frac{1}{1+\beta} \left[y_0 + \frac{y_1}{1+r} \right] - \frac{y_1}{1+r} = \frac{1}{1+\beta} y_0 + \frac{y_1}{1+r} \left[\frac{1}{1+\beta} - 1 \right] \\ &= \frac{1}{1+\beta} y_0 - \frac{\beta}{1+\beta} \frac{y_1}{1+r} \end{aligned}$$

Note that the assumption that $y_0 > \frac{y_1}{\beta(1+r)}$ implies that $y_0 > \frac{\beta y_1}{(1+r)}$ (because $0 < \beta < 1$). Therefore, this term is positive, and George is a borrower.

(d)(i) It will hold for the person that lends (Jerry), but not for the person who borrows (George). The reason is that the optimal plan, as governed by the Euler equation, is no longer satisfying all the constraints. In other words, now there is a borrowing constraint that will be binding. Note that this is only relevant for the consumer who was borrowing before. The person who would like to lend is not constrained, and so his consumption plan would still obey the Euler equation.

(d)(ii) It will be better to give the rebate to George, as he is more likely to consume all or a larger fraction of the transfer. The reason is that borrowing constraint implies that he would like to consume more but can't. Therefore, giving him the rebate results in him increasing current consumption. On the other hand, Jerry will smooth according to the Euler equation, and so while his current consumption may increase, it will certainly be at a lower rate.